

09. INFLUENCE OF EMBRYONIC FOLDING ON THE UPPER DIGESTIVE TRACT

DURATION : 05:28

STUDY SHEET

COURSE SUMMARY

This video delves into the crucial impact of embryonic folding on the development of the upper digestive tract. It illustrates how the flexion of the embryo influences the mesenchyme and creates the base of the skull, while highlighting the organisation of structures such as the neural tube, heart, and vascular system. Key terms such as the amniotic cavity, notochord, and foregut are discussed to demonstrate their role in the formation of the digestive tract, thus linking organ development to the vital dynamics of the amniotic fluid. This is both a theoretical and practical approach, essential for any therapist wishing to understand the subtleties of embryonic development.

INFLUENCE OF EMBRYONIC FOLDING ON THE UPPER DIGESTIVE TRACT

The **importance of embryonic folding** is manifested in the upper part of the intestine. This **flexion** movement of the embryo leads to significant changes in its development.

During folding, the **mesenchyme**, which is the mesenchymal tissue beneath the brain vesicles, becomes perpendicular to the embryo's longitudinal axis. This movement creates a beautiful curvature, leading to **compression** between the front and back, which contributes to the formation of the skull base. This dynamic process generates a **cartilaginous field** that, initially loose, gradually densifies to form compressed **primitive cartilages**, thereby releasing significant energy.

The **amniotic cavity** and the **vitelline cavity** play a crucial role in this development. As the embryo folds, the vitelline cavity shifts, and the neural tube begins to coil. The cardiac area, initially anterior, is also influenced by this flexion movement.

The study of this movement proceeds from the periphery inwards, from the visible face, such as the mouth, to the pharyngeal intestine. This flexion movement organises the **peculiarity of the face**. The **notochord** induces the neural tube and arteries, while the coiling around the arterial system organises itself.

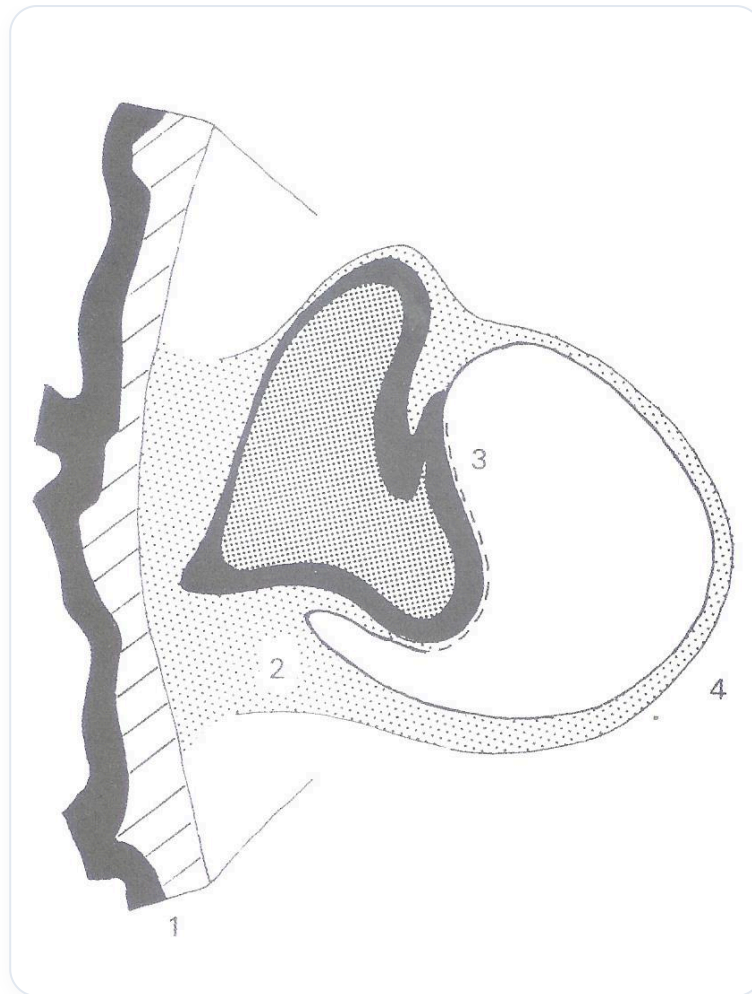


FIG. — Taste bud

The upper part of the intestine forms as an **articulation** between the heart and the brain. The **foregut** appears, representing the developing upper intestine. This articulation is oriented and organised in space-time by the **amniotic fluid**, with the vascular system as a fulcrum.

The heart, visible in an elevated position, and the coiling of embryonic structures illustrate this articulation. The further development progresses, the stronger the organisation becomes. The large vitelline sac and its remnant form, while the upper part organises itself through flexion movements.

The **amniotic cavity** and the heart create a space that becomes an articulation between the heart and the brain, propelled by amniotic growth. This process marks the beginning of the formation of the digestive tract, which organises itself on a longitudinal plane. The delimitation of the upper part of the intestine begins, integrating the **allantois** and the **cloacal membrane**.

 Explanatory diagram

FIG. — Explanatory diagram

Between this membrane and the pharyngeal membrane, the mouths at the buccal level develop. This articulation between the brain, the amniotic sac, and the heart is organised by the differential growth rate of the vascular system, establishing a balance between the brain and the heart, oriented by the **mesoderm**.

The appearance of a vessel depends on various components: a field, a metabolic concentration, a trajectory, **vacuolations**, and the presence of interstitial particles. These elements are dynamic, which explains why some vessels may appear and disappear during embryonic development. Initially, two large aortas centralise into a single aorta.

The influence of this folding is therefore essential for organising the **internal delimitation** of the digestive tract. We will now delve into the details of this process.

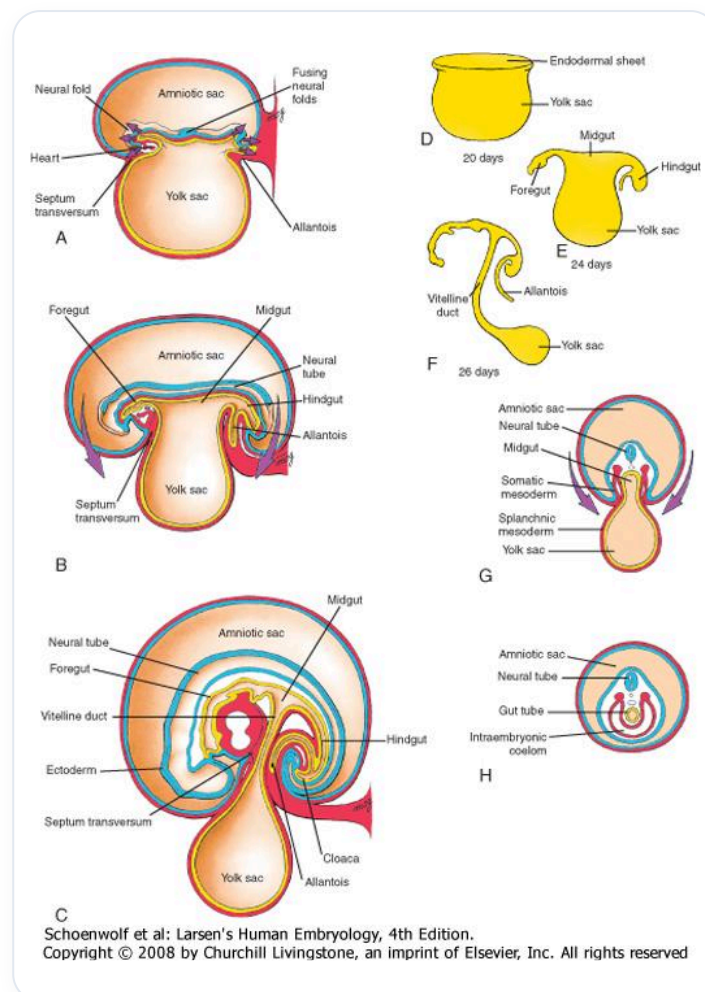


FIG. — Embryonic digestive tract

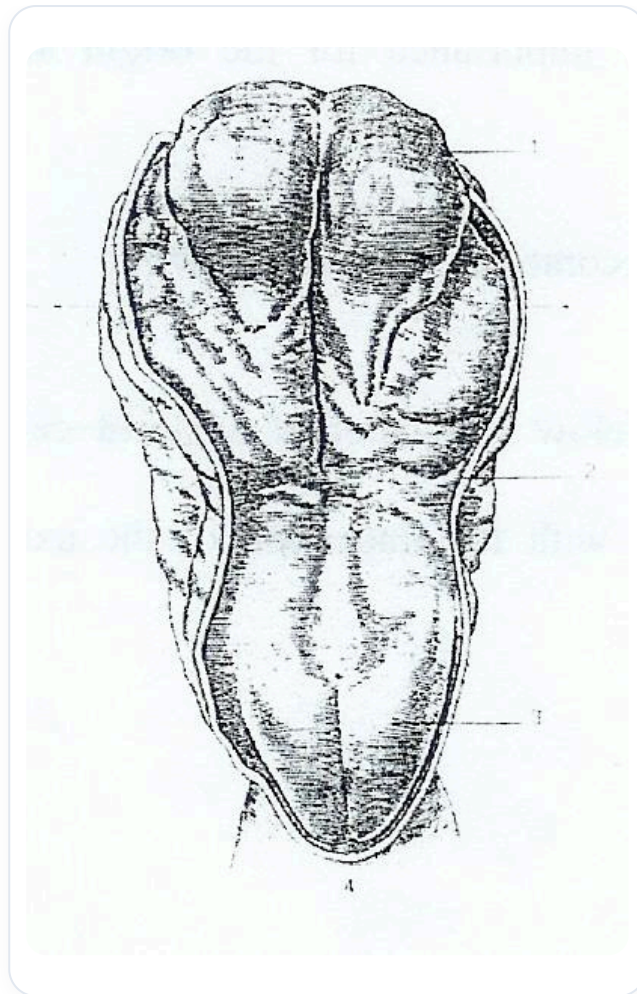


FIG. — *Human embryonic brain*

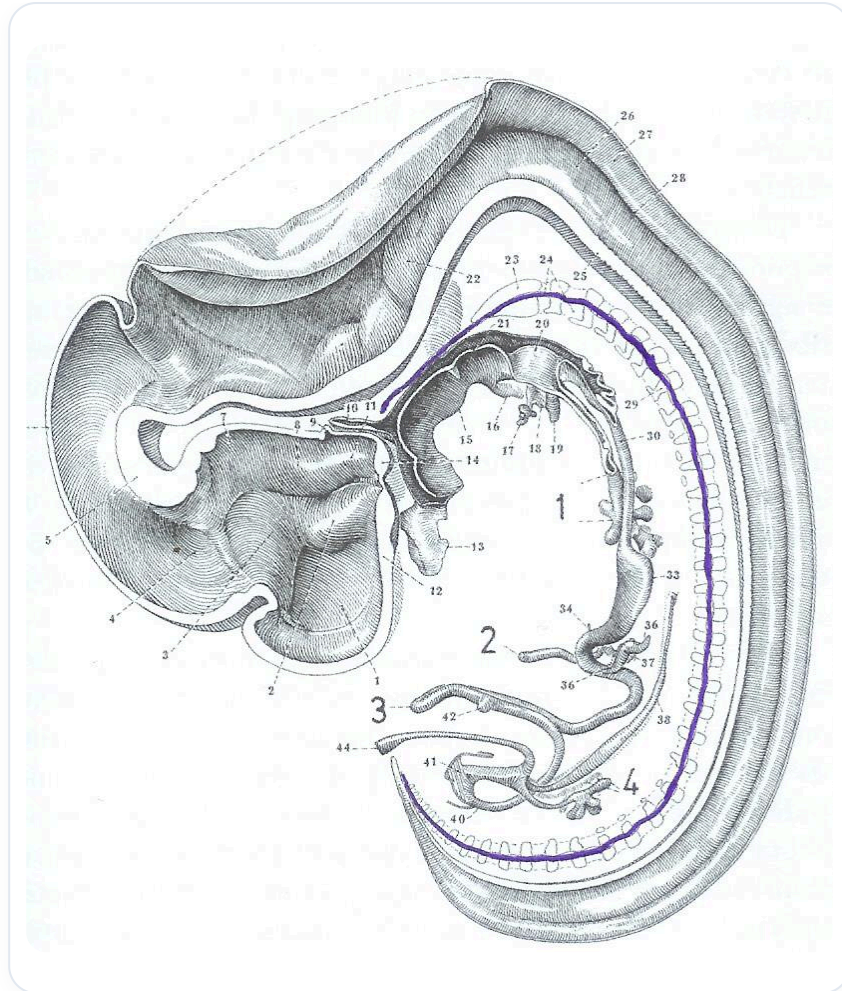


FIG. — Human embryo sagittal

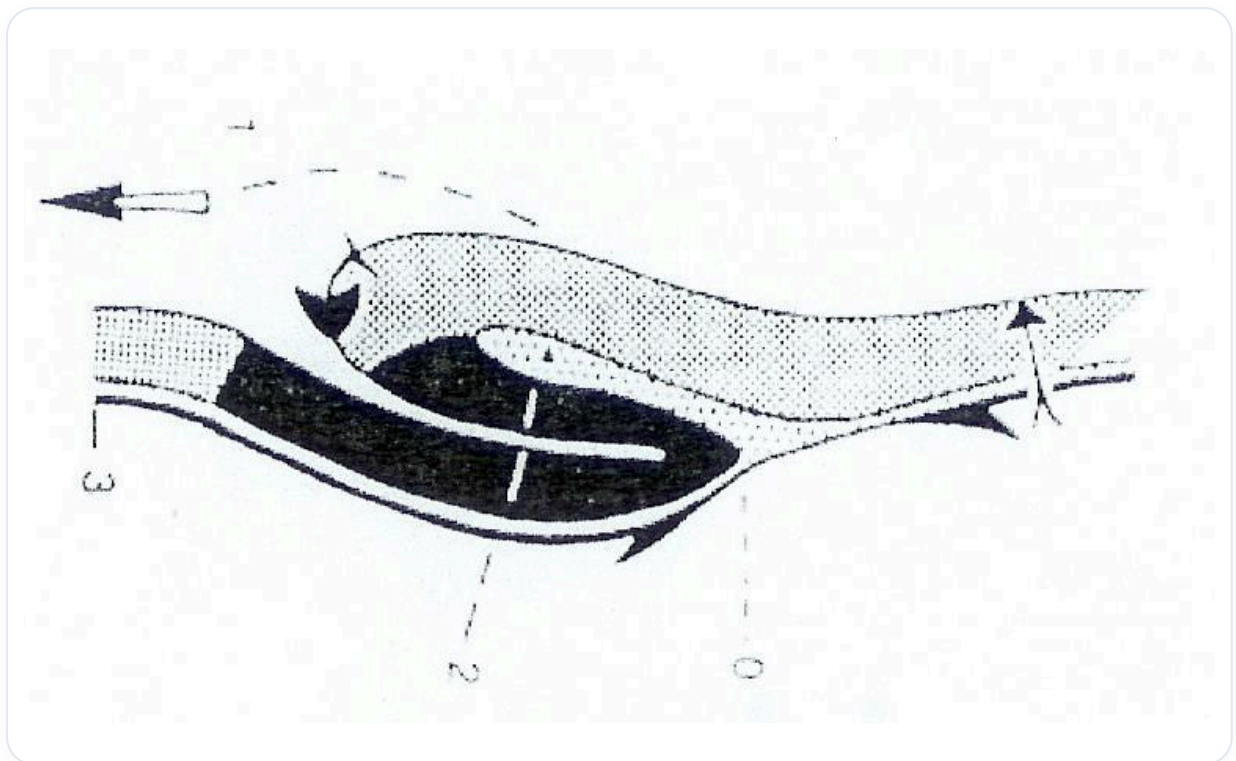


FIG. — Formation of the notochord

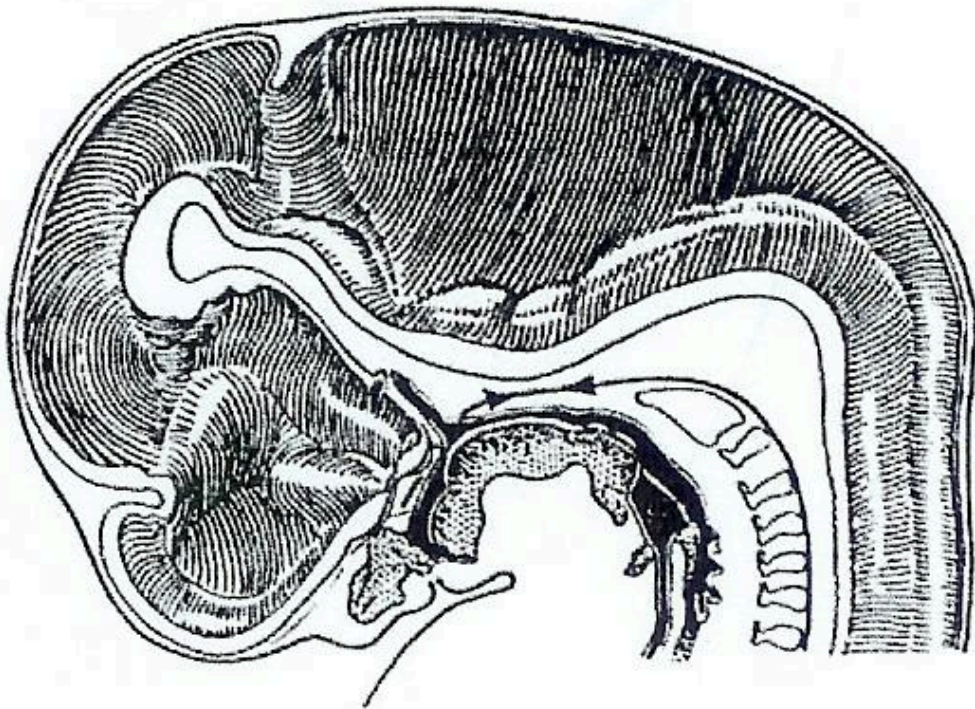


FIG. — Sagittal human embryo

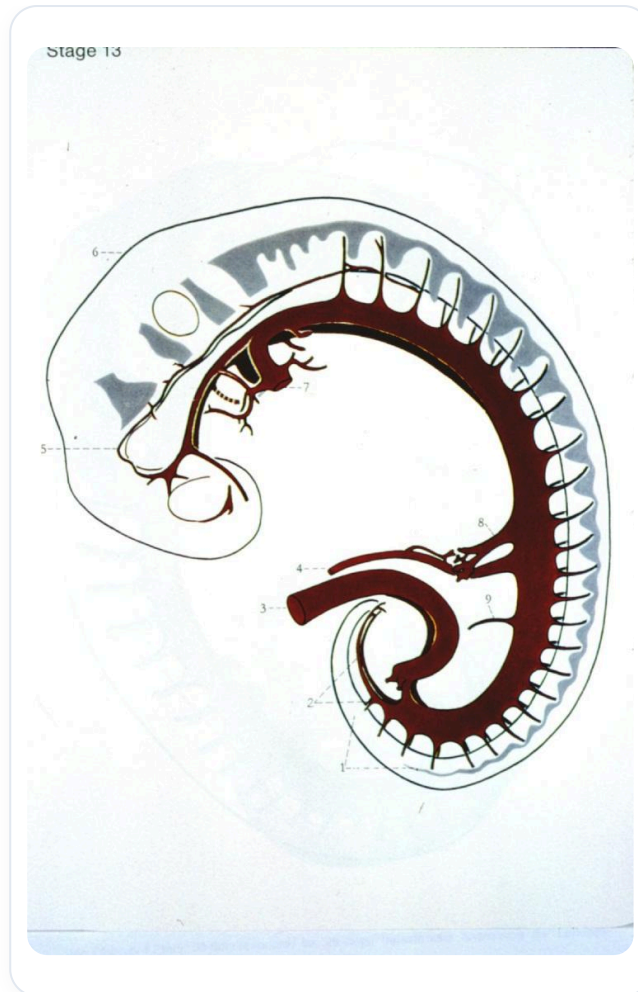


FIG. — *Vascularized embryo*

 Explanatory diagram

FIG. — *Explanatory diagram*